cBioPortal
FOR CANCER GENOMICS

Tutorial 1
Single Study Exploration

Explore all data in a dataset

Last update: November 3, 2026

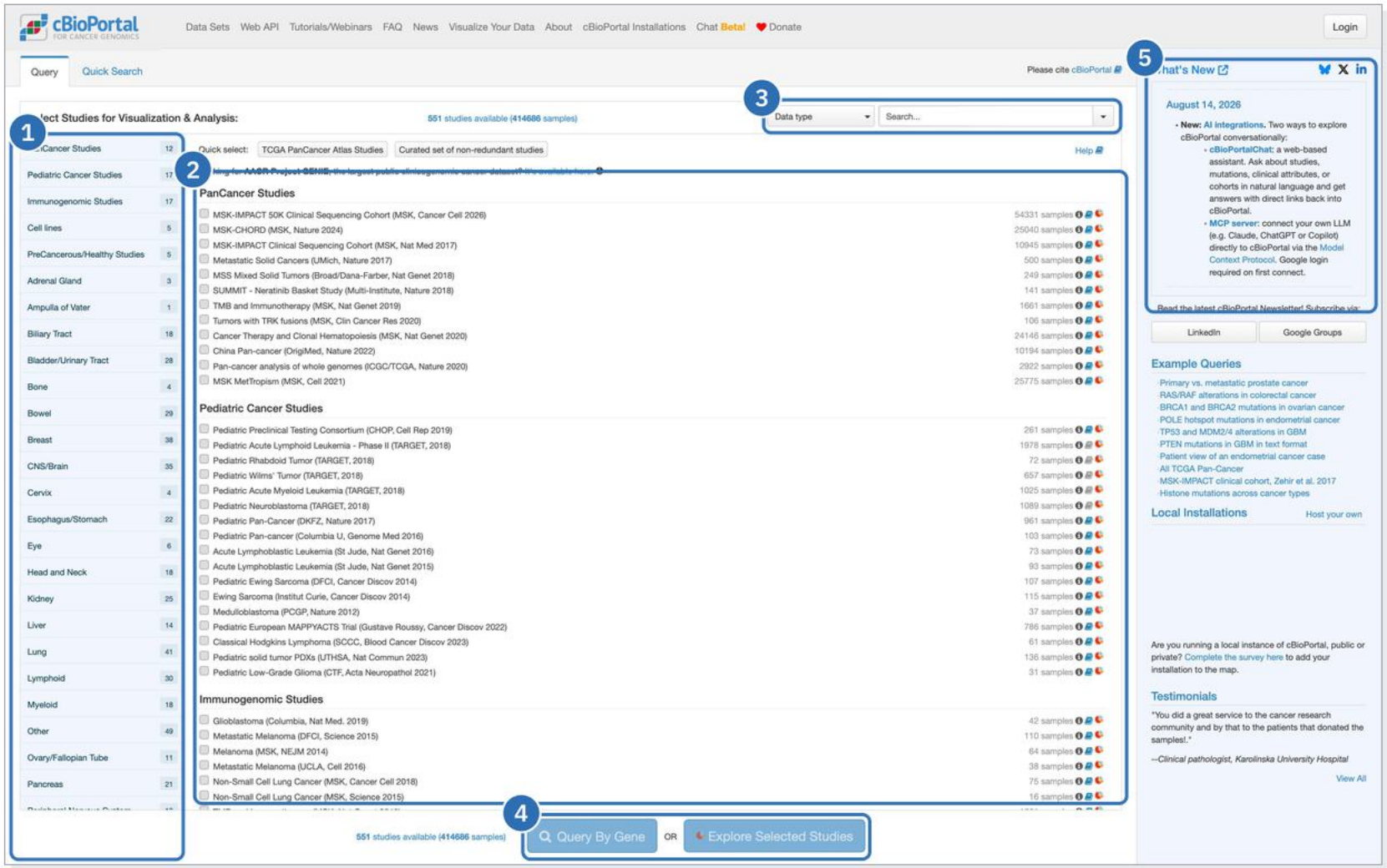
cbioportal.org/tutorials

Tutorial objectives

- Introduce the cBioPortal main page
- Find a study on the main page: search by name, or filter by data type
- Walk through the study view tabs: Summary, Clinical Data, CN Segments, Plots and Slide Microscopy
- Add charts, select subsets of samples, create groups and share a virtual study
- Run a query from the study view

Example study throughout: Brain Lower Grade Glioma (TCGA, PanCancer Atlas), 514 patients and samples.

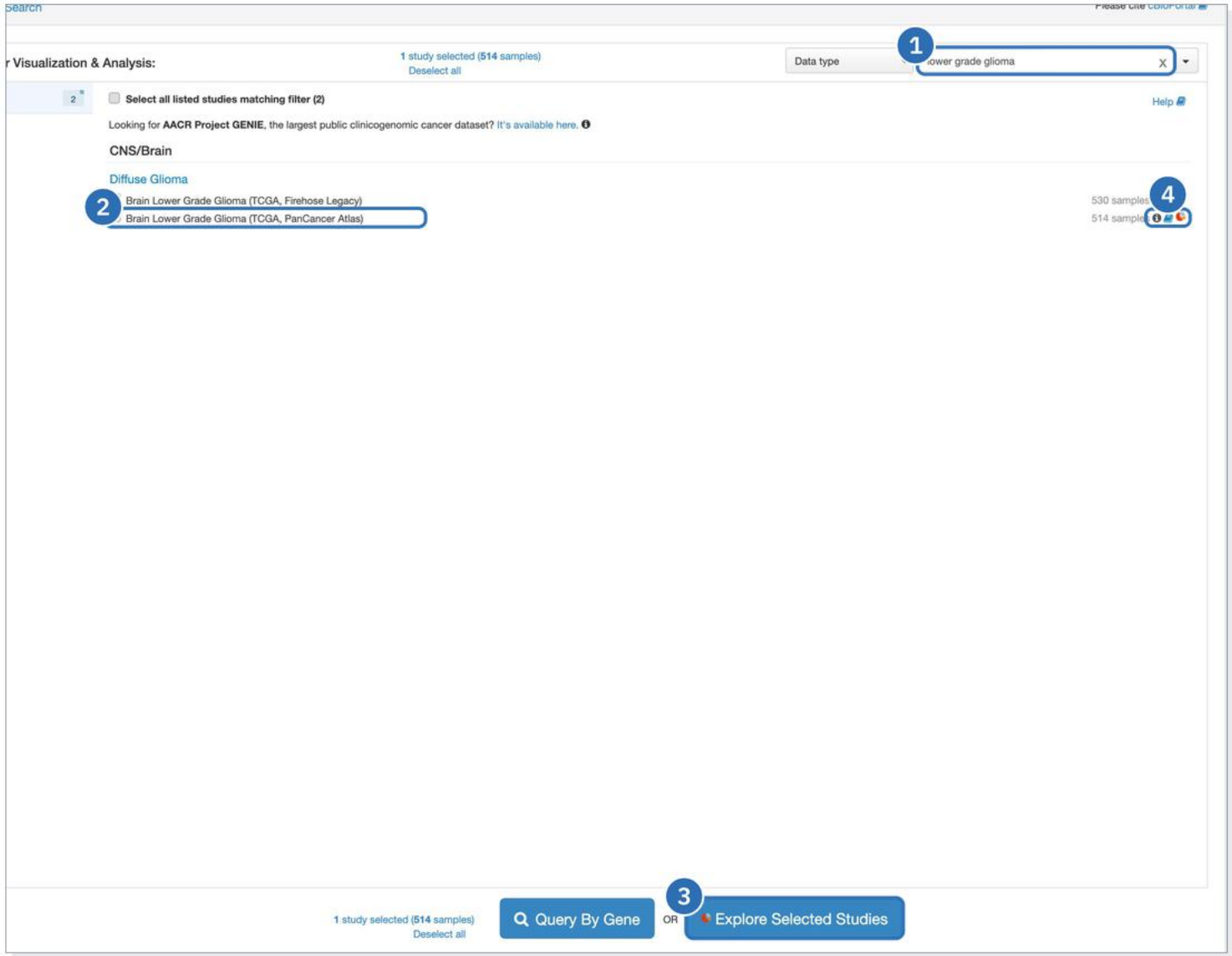
cBioPortal main page



1. Organ systems, with the number of studies in each. Click one to filter the list.
2. All 551 studies (414,686 samples), grouped by organ system. Quick select picks curated sets.
3. Search studies by name, or filter by the data types they contain.
4. With studies ticked: Query By Gene, or Explore Selected Studies.
5. What's New: cBioPortal Chat and the MCP server answer questions in plain language.

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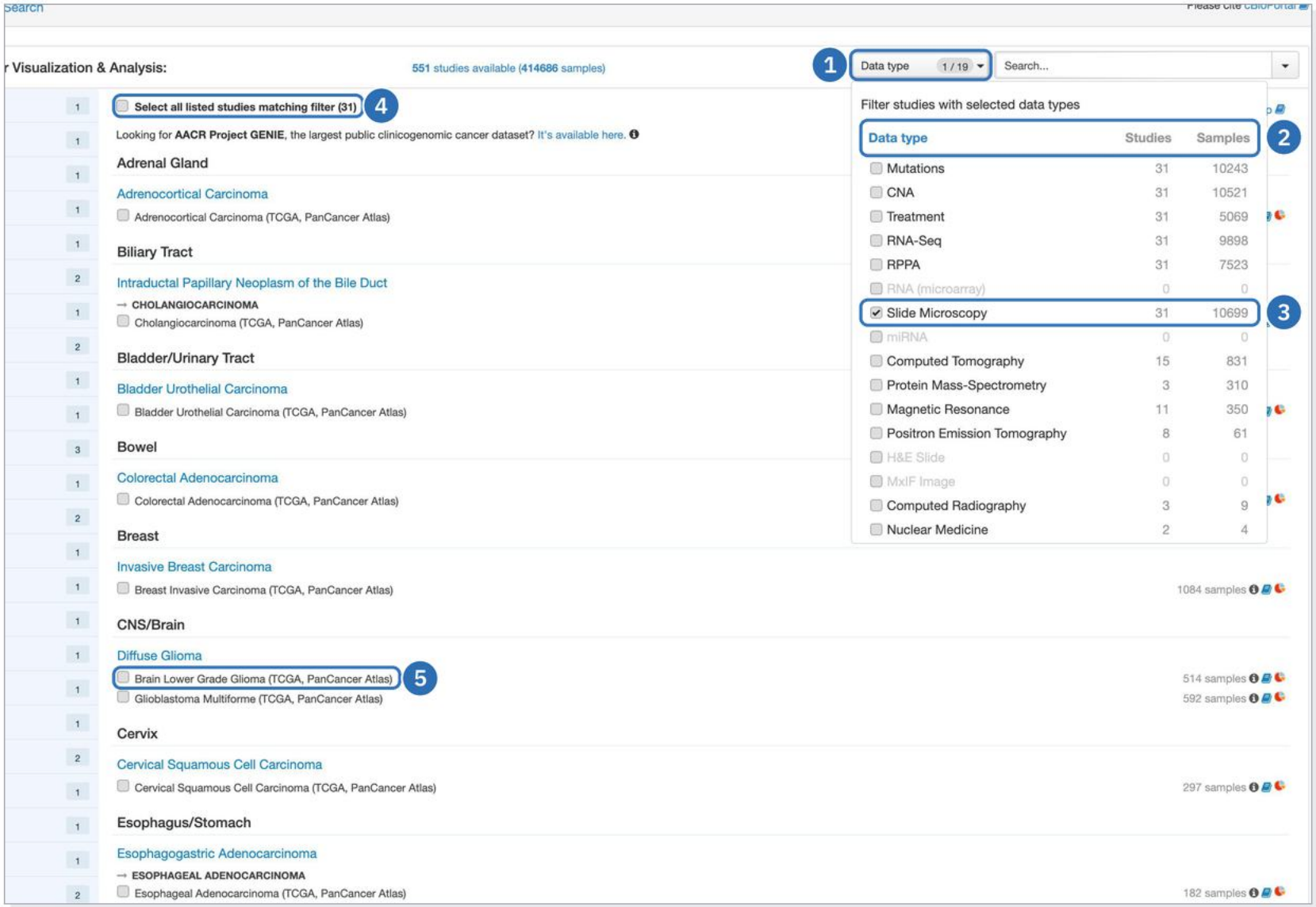
Selecting a study: search by name



1. Filter the list: type in the search box (here, “lower grade glioma”).
2. Tick the study. The counter shows 1 study selected (514 samples).
3. Click Explore Selected Studies to open the study view.
4. Or click the pie icon next to a study to open its summary directly.

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Selecting a study: filter by data type

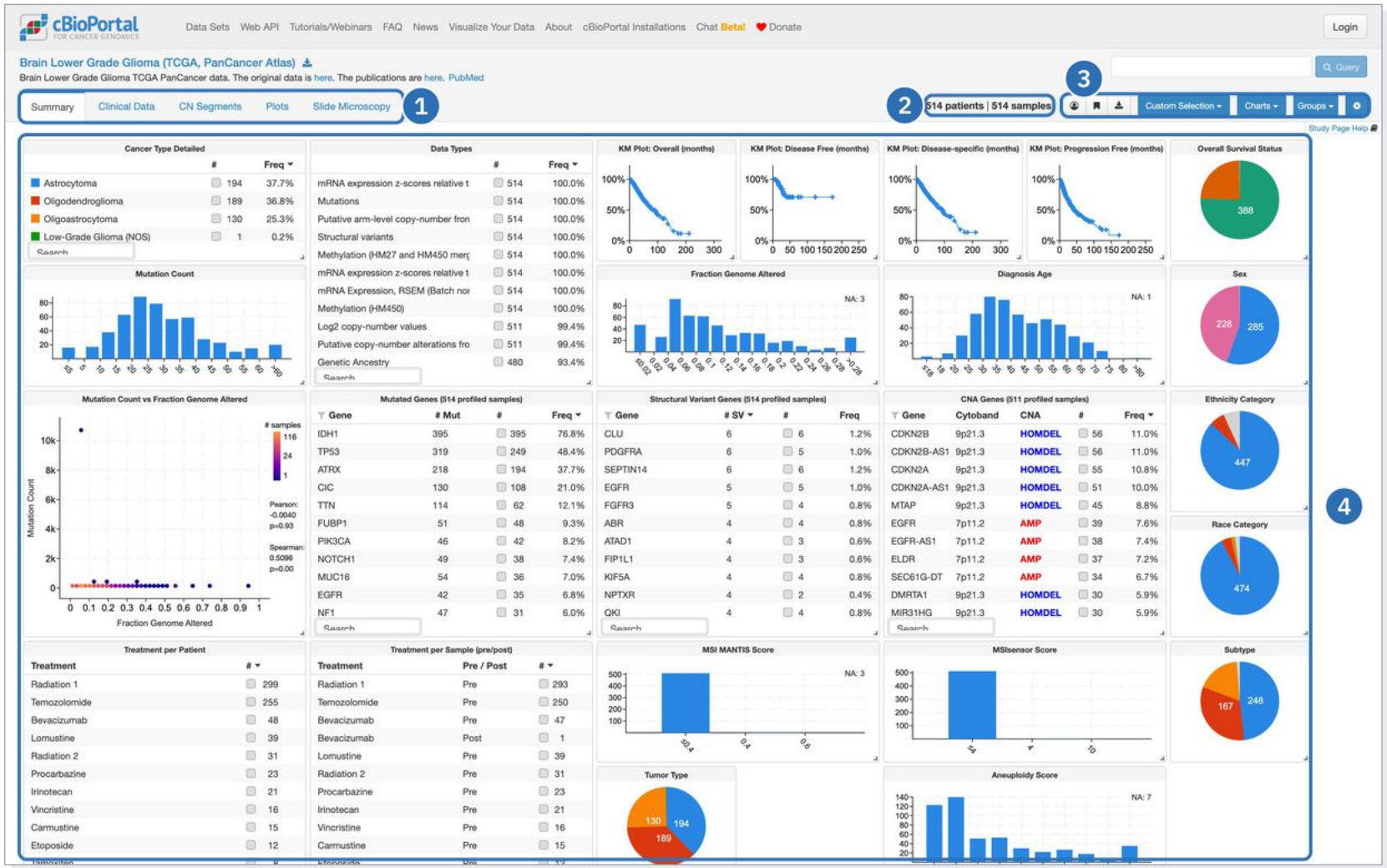


1. Or narrow the list by the data you need: open the Data type menu.
2. Each data type shows how many studies and samples have it.
3. Tick one or more. Here: Slide Microscopy, available for 31 studies (10,699 samples).
4. The list now shows only matching studies; you can select them all at once.
5. Our example study is among them: tick it and click Explore Selected Studies.

Once you select a study

- Click Explore Selected Studies (or the pie icon next to a study) to open the study view on its Summary tab.
- Here you can explore features of the samples, such as mutated genes or sex, for the whole study or for a subset, and start a query.
- The charts and tabs you see depend on the data available for the study you selected.

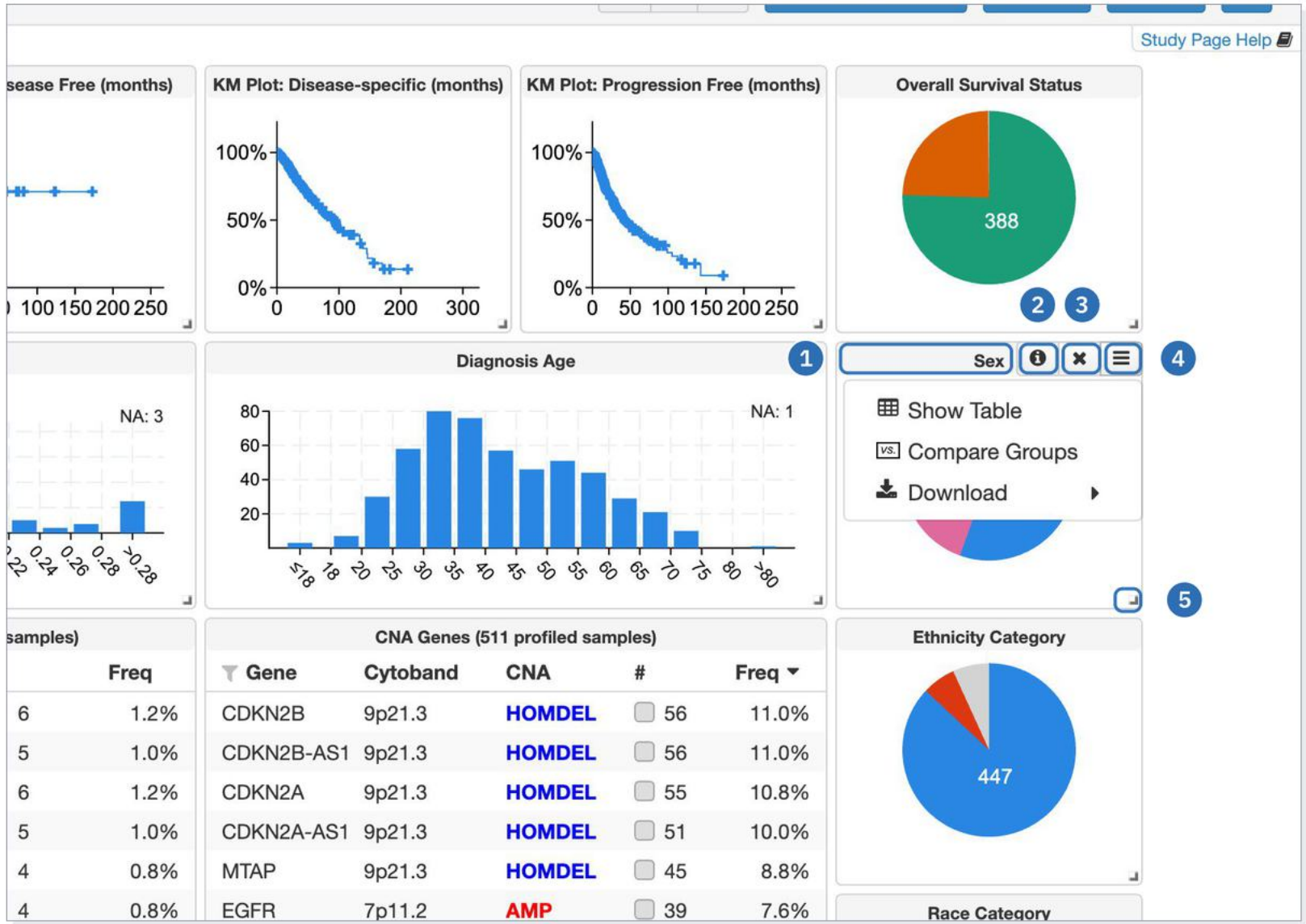
Study view: Summary tab



1. Tabs: Summary, Clinical Data, CN Segments, Plots, Slide Microscopy. They depend on the study's data.
2. The current selection: 514 patients, 514 samples.
3. View the selected cases, share a virtual study, download; Custom Selection, Charts, Groups.
4. Every chart summarizes one attribute. Click a chart to select samples; all charts update.

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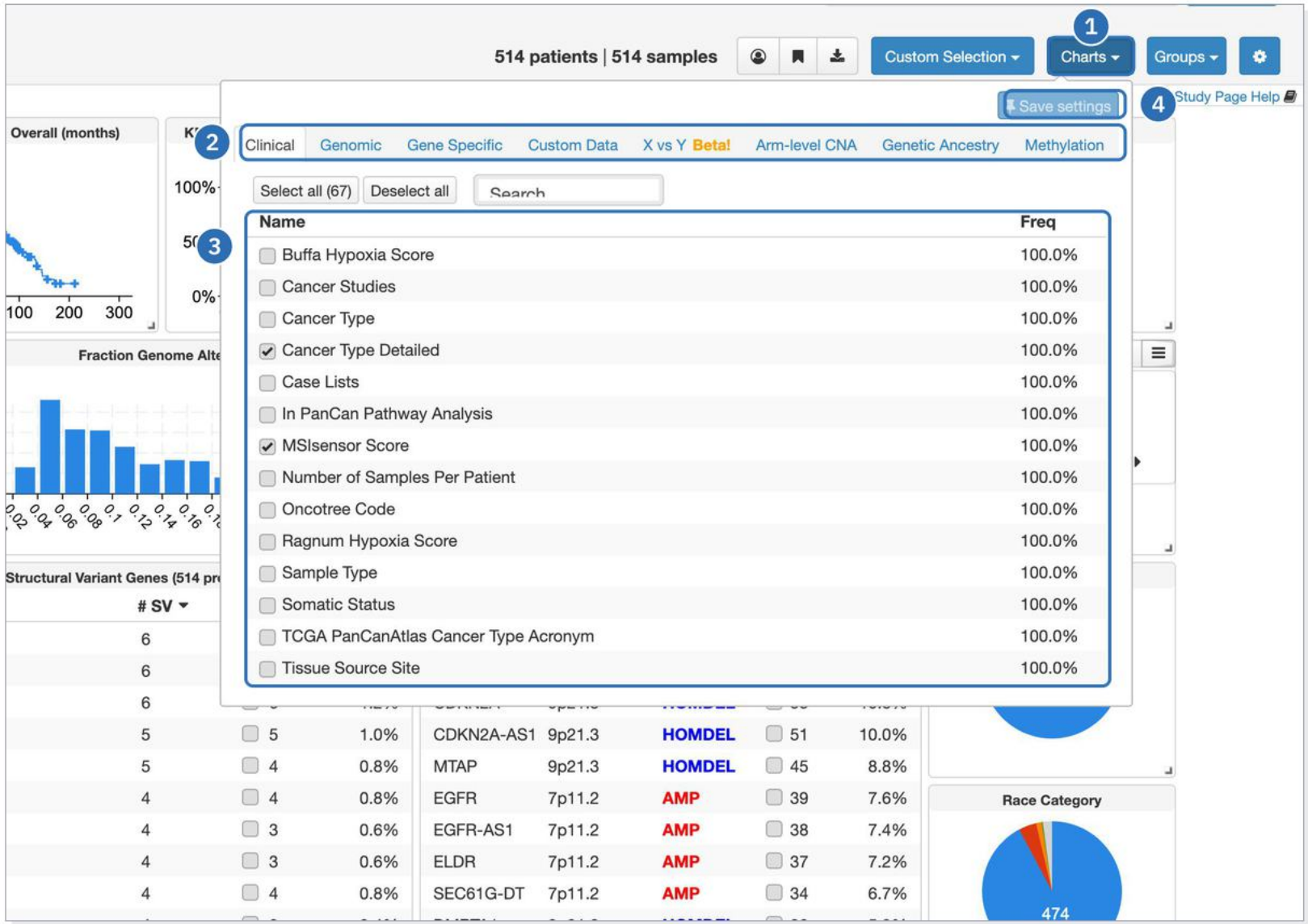
Summary tab: working with charts



1. Hover over a chart's title bar to show its buttons. Drag the title bar to move the chart.
2. The i button describes the data.
3. The x button removes the chart.
4. Hover over the menu icon for Show Table, Compare Groups (one group per value, see the Group Comparison tutorial) and Download.
5. Drag the corner to resize the chart.

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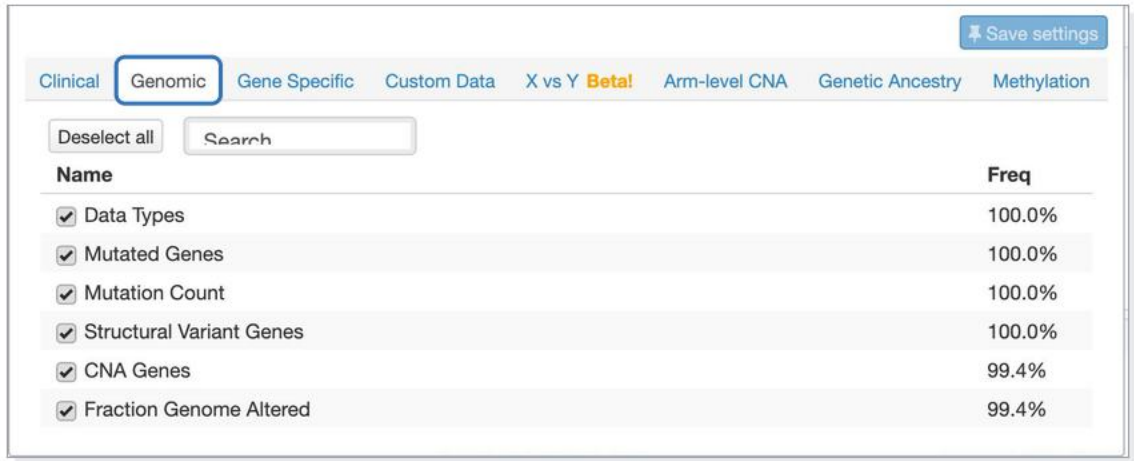
Summary tab: adding charts



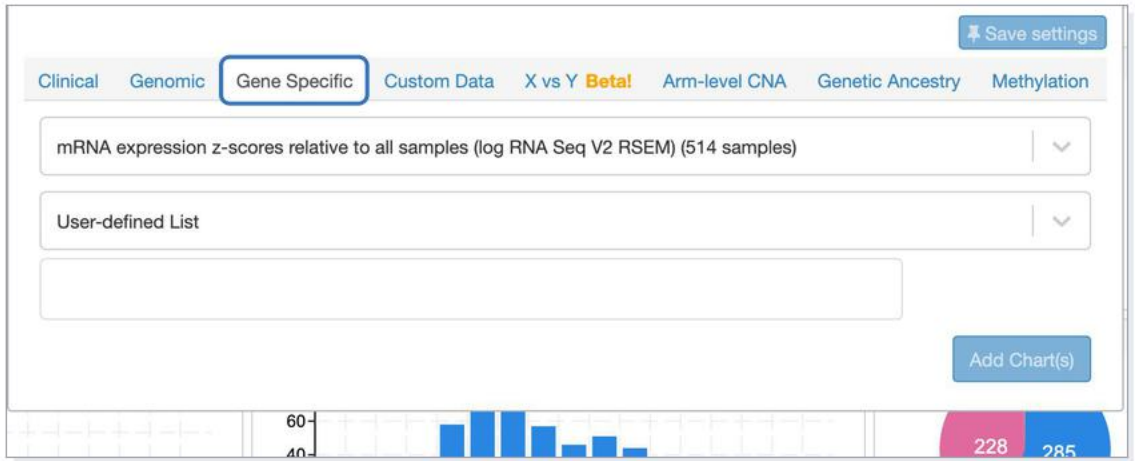
1. The Charts menu adds charts. They work like any other: filter, compare, download.
2. Clinical, Genomic, Gene Specific (e.g. mRNA of one gene), Custom Data (your own groups), X vs Y (Beta), and study-specific tabs such as Arm-level CNA, Genetic Ancestry, Methylation.
3. Tick attributes to show them. Freq is the share of samples with data.
4. Save settings keeps this chart layout.

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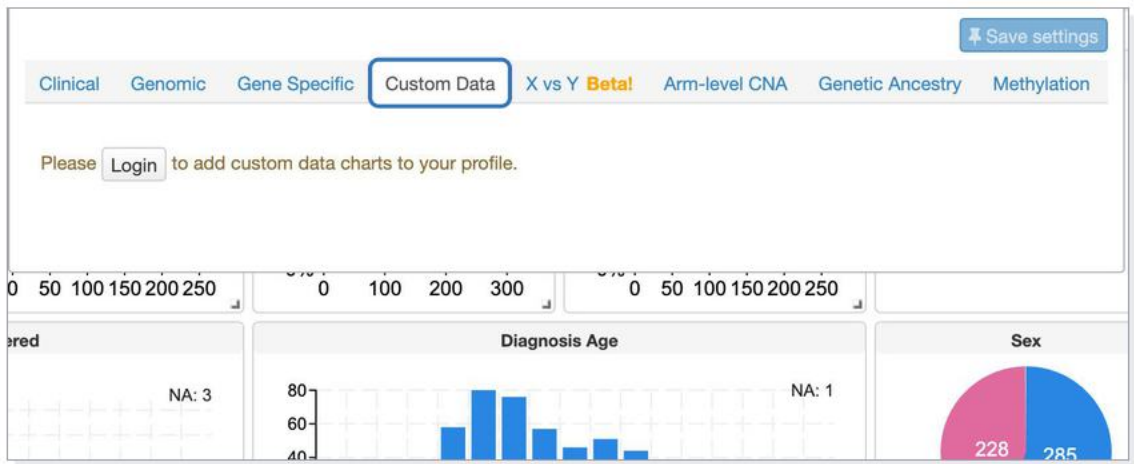
Summary tab: more kinds of charts



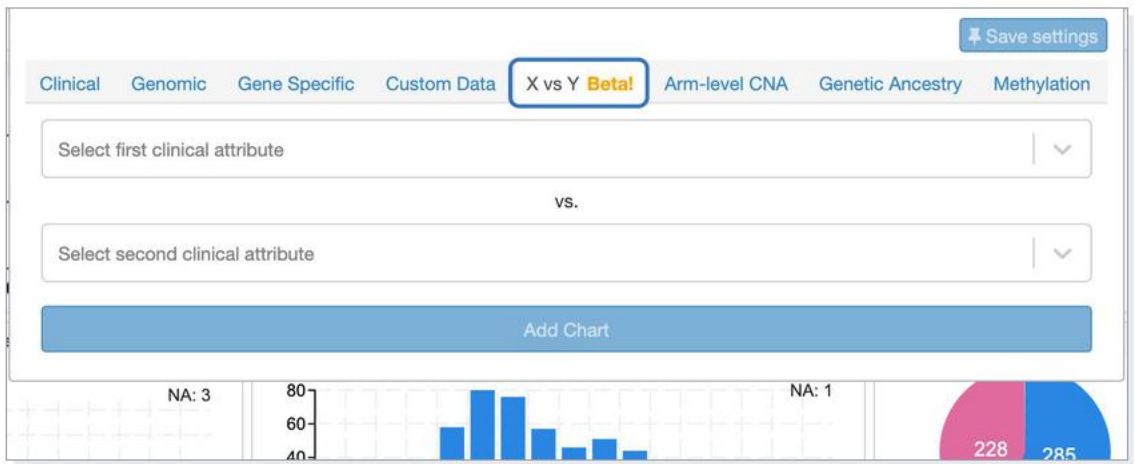
Genomic: charts that summarize the genomic data (mutated genes, CNA genes, mutation count, ...).



Gene Specific: a chart for one gene from any profile with continuous data, e.g. mRNA expression.

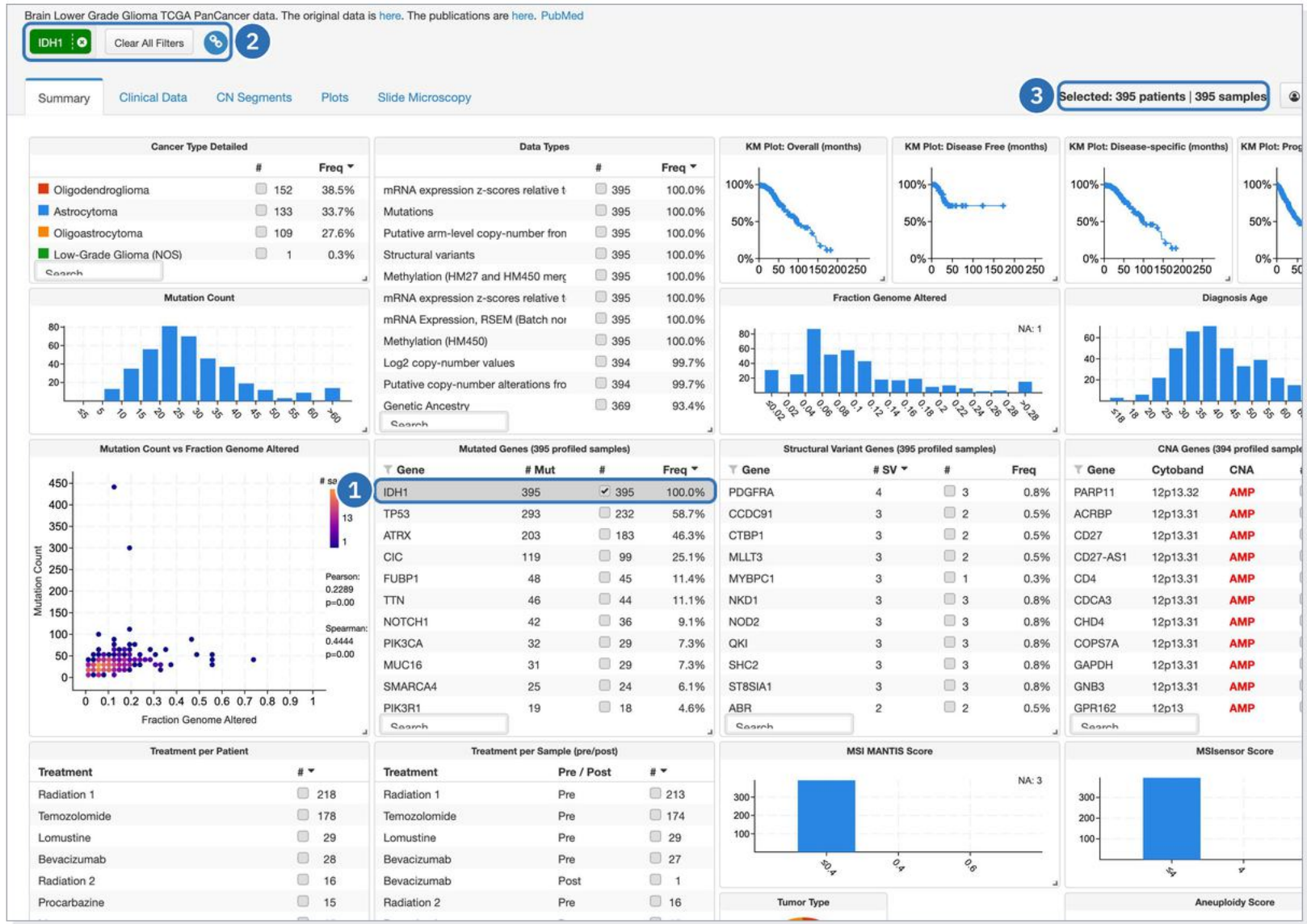


Custom Data: your own values for the samples, e.g. groups from your own analysis (requires login).



X vs Y (Beta): a chart of two clinical attributes. Arm-level CNA, Genetic Ancestry and Methylation appear for some studies.

Summary tab: selecting a subset of samples



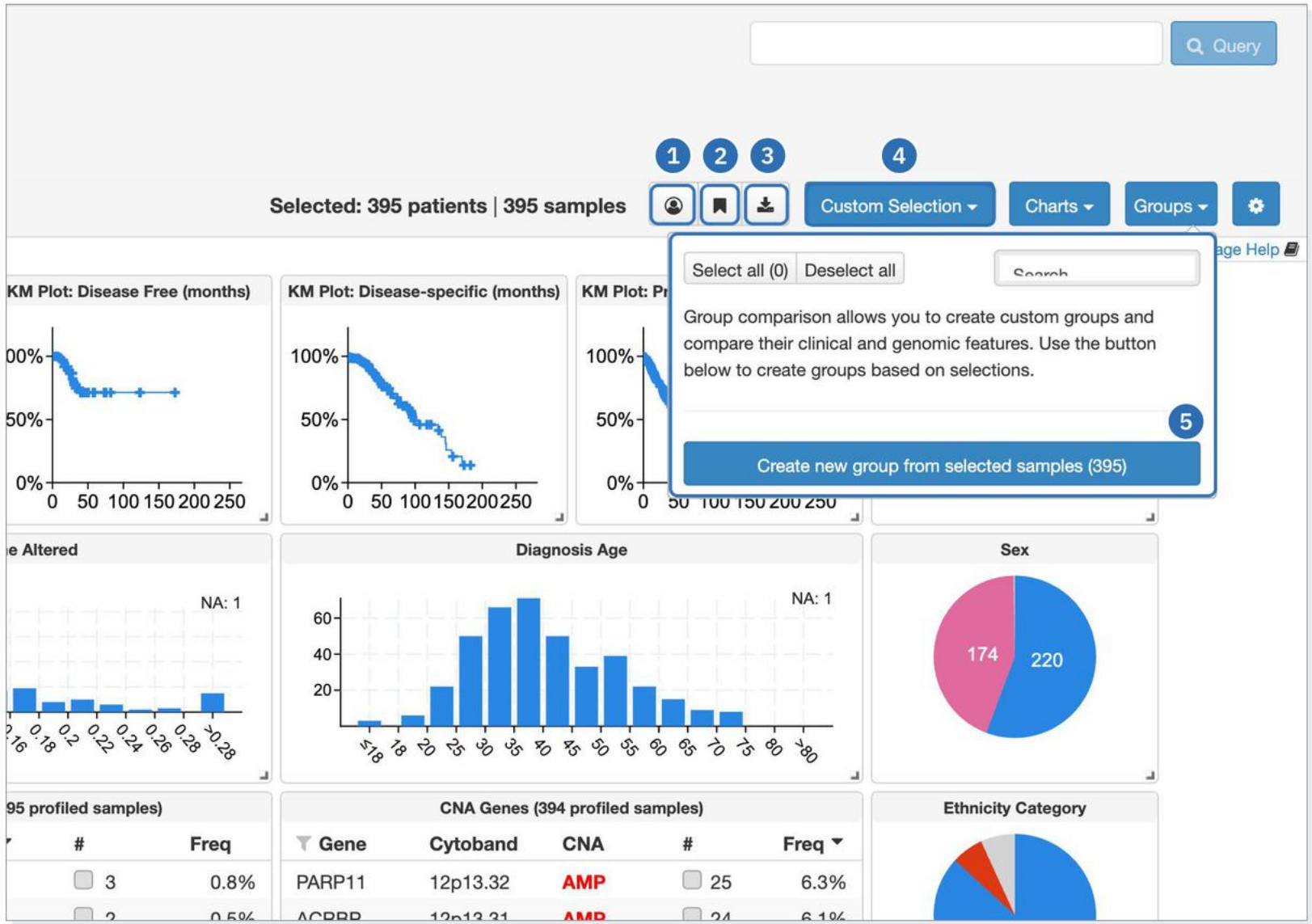
1. Use any chart to select samples. Here: tick IDH1 in Mutated Genes, then Select Samples.

2. Active filters are listed here. Click x on a filter to remove it, or Clear All Filters. The link button copies a shareable link with the filters applied.

3. All charts update: 395 IDH1-mutant samples. Add more filters to narrow down.

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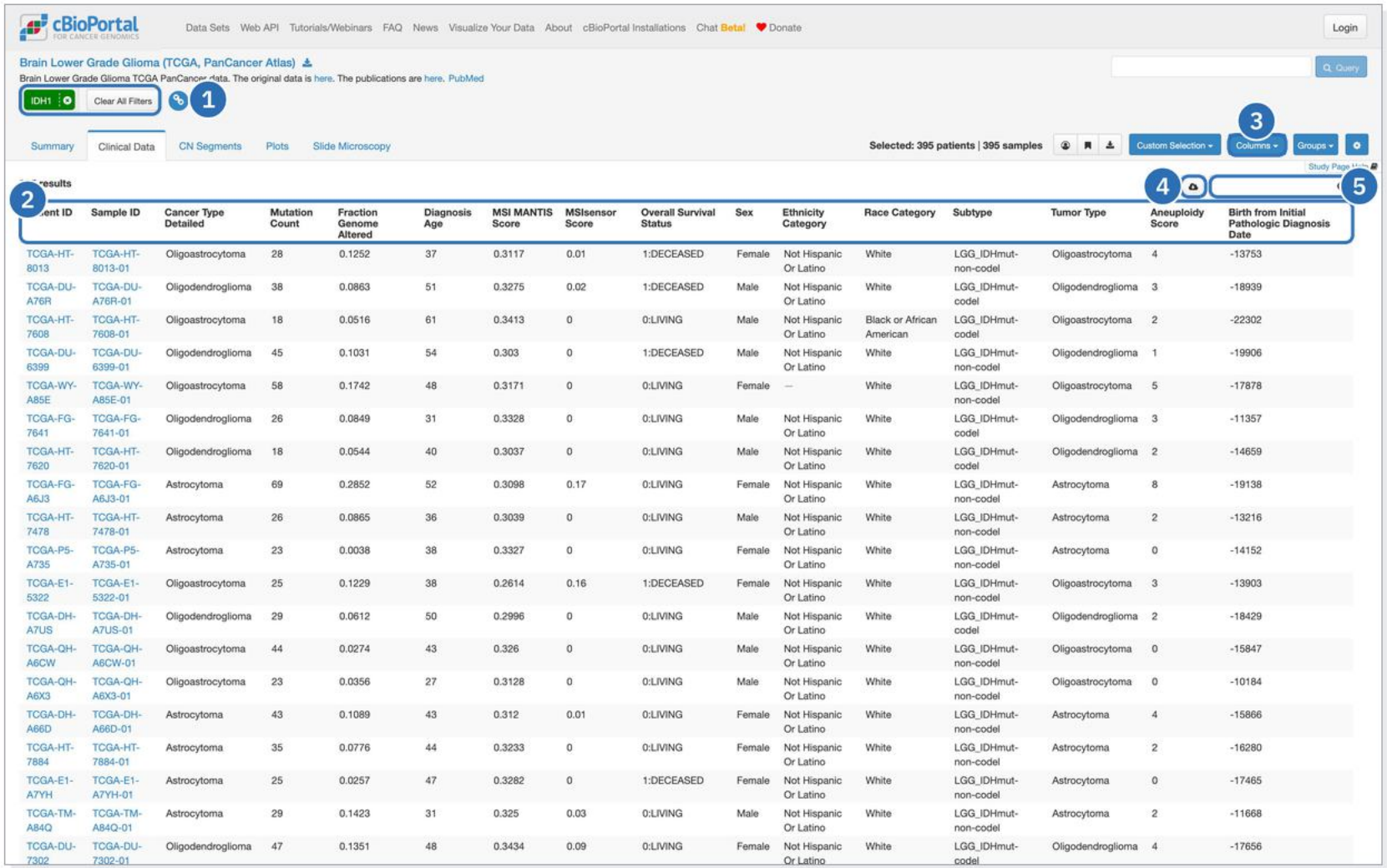
Study view: groups, custom selection and sharing



1. View all selected cases in Patient View (see the Patient View tutorial).
2. Share Virtual Study: save the selection as a virtual study (see the Virtual Studies tutorial).
3. Download clinical data for the selected cases.
4. Custom Selection: paste a list of sample or patient IDs to filter to.
5. Groups: create a group from the selected samples and compare groups (see the Group Comparison tutorial).

Link to this page

Clinical Data tab



1. Filters from the Summary tab apply here: 395 results.
2. Click a column header to sort. Scroll right for more columns.
3. The Columns menu shows more attributes (they vary by study).
4. Download the table.
5. Search the table.

Link to this page

CN Segments tab

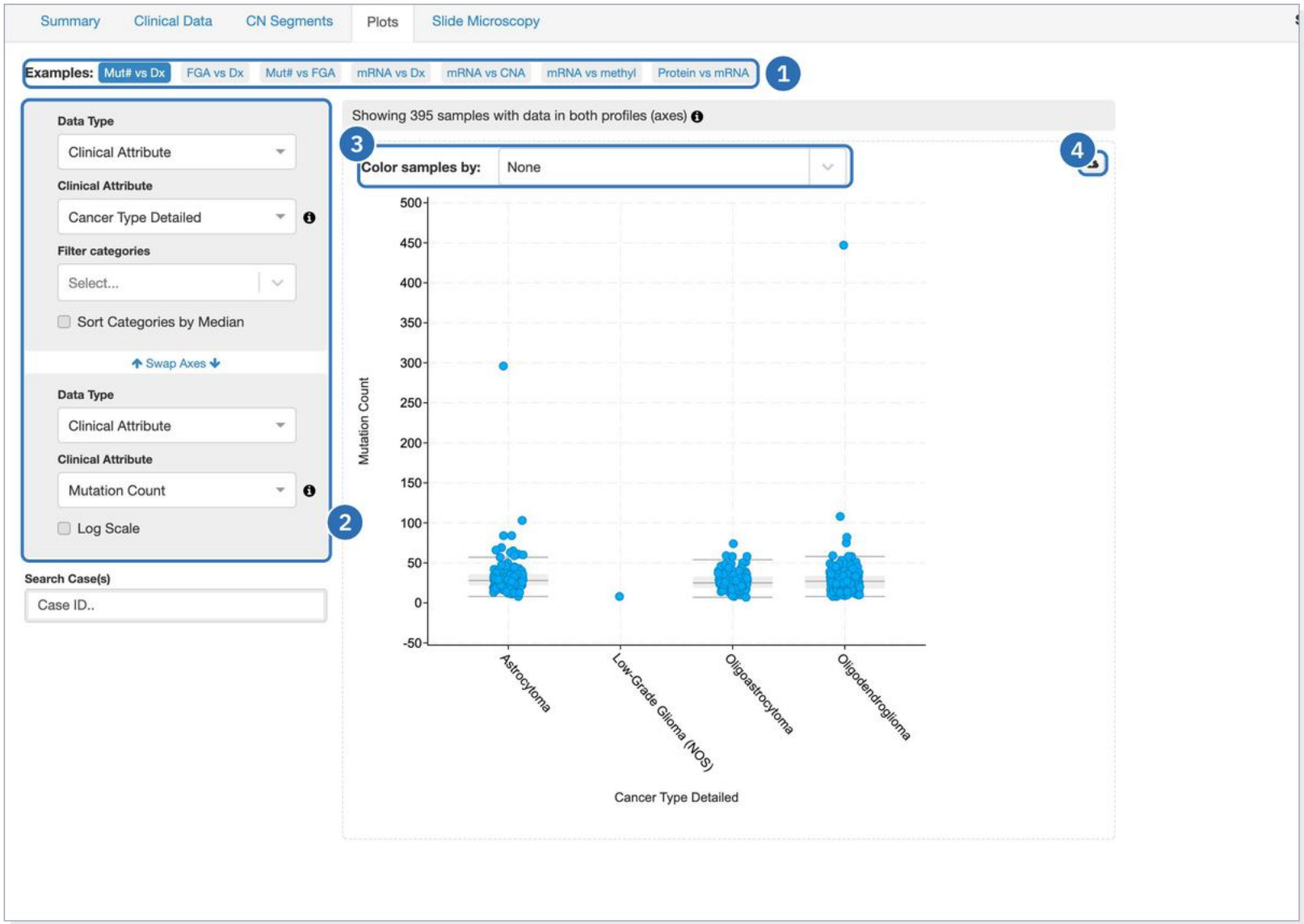


IGV view: 2021 capture of the same selection

1. Enter a gene or genomic coordinates to zoom in. Filters from the Summary tab apply.
2. When copy-number data is available, the Integrative Genomics Viewer (IGV) shows segments across the genome, one row per sample.

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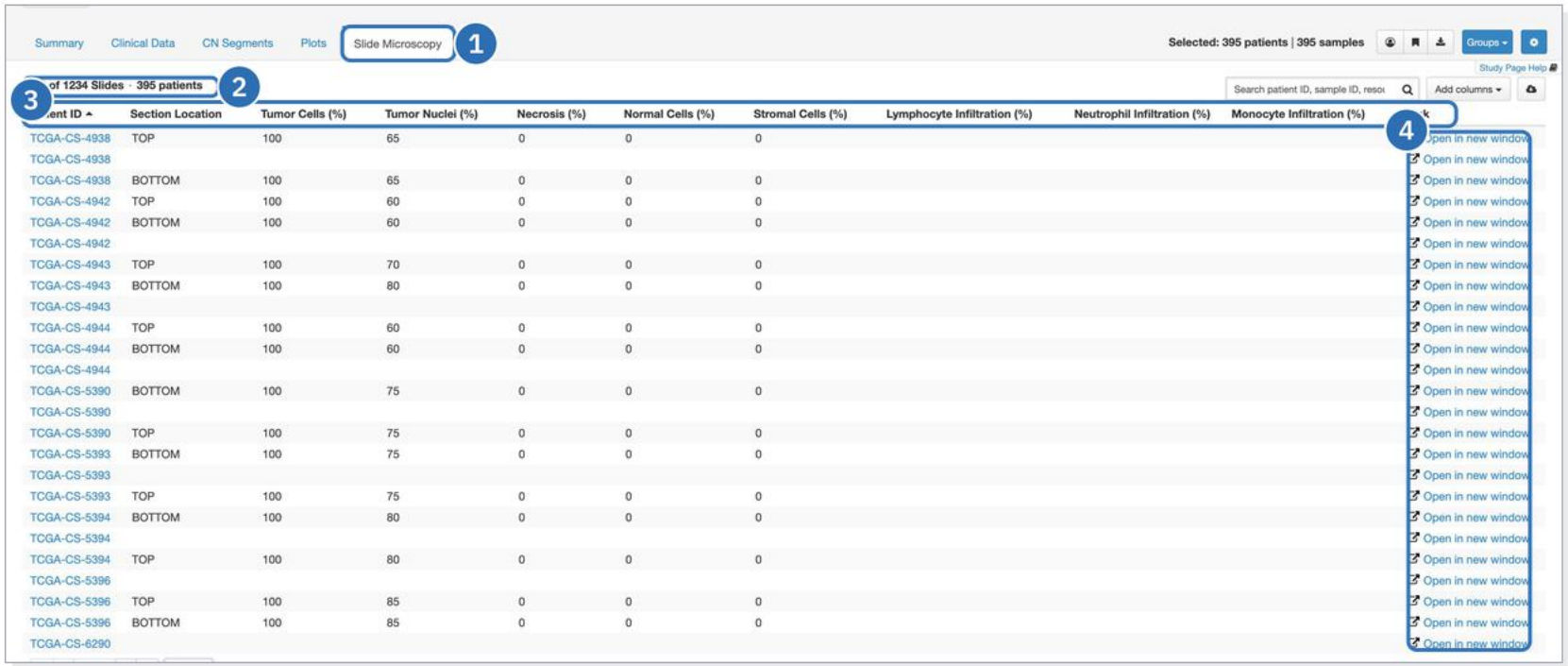
Plots tab



1. Start from an example, such as mutation count by cancer type (Mut# vs Dx).
2. Or choose the data type and attribute for each axis: clinical attributes, mutations, copy number, expression and more.
3. Color samples by another attribute.
4. Download the plot or its data.

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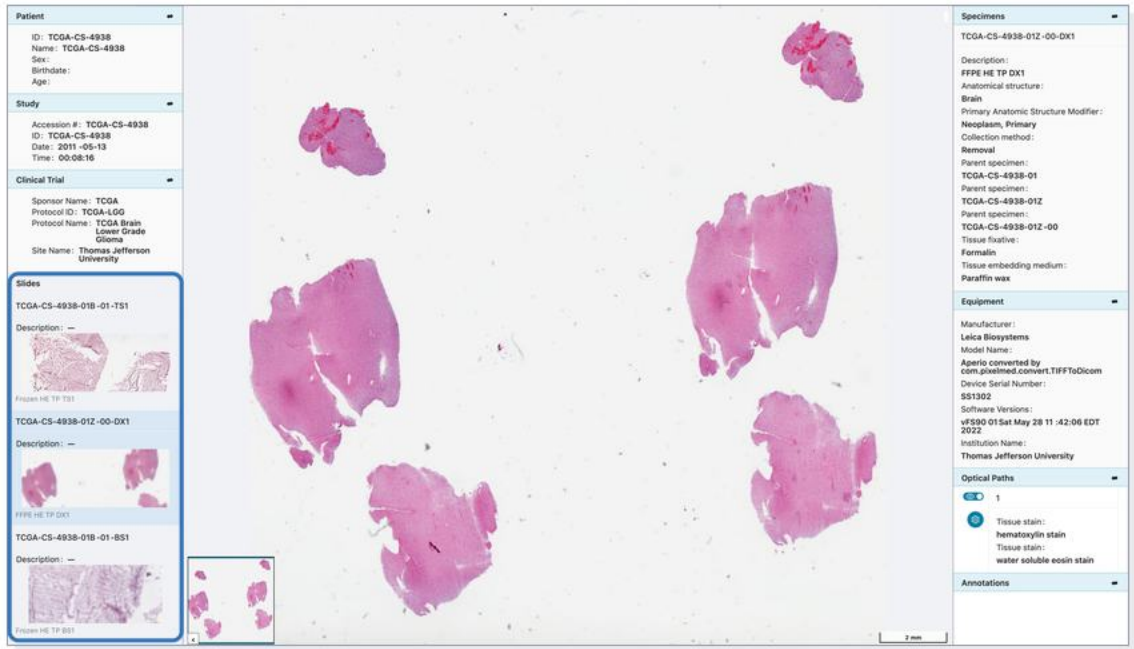
Slide Microscopy tab



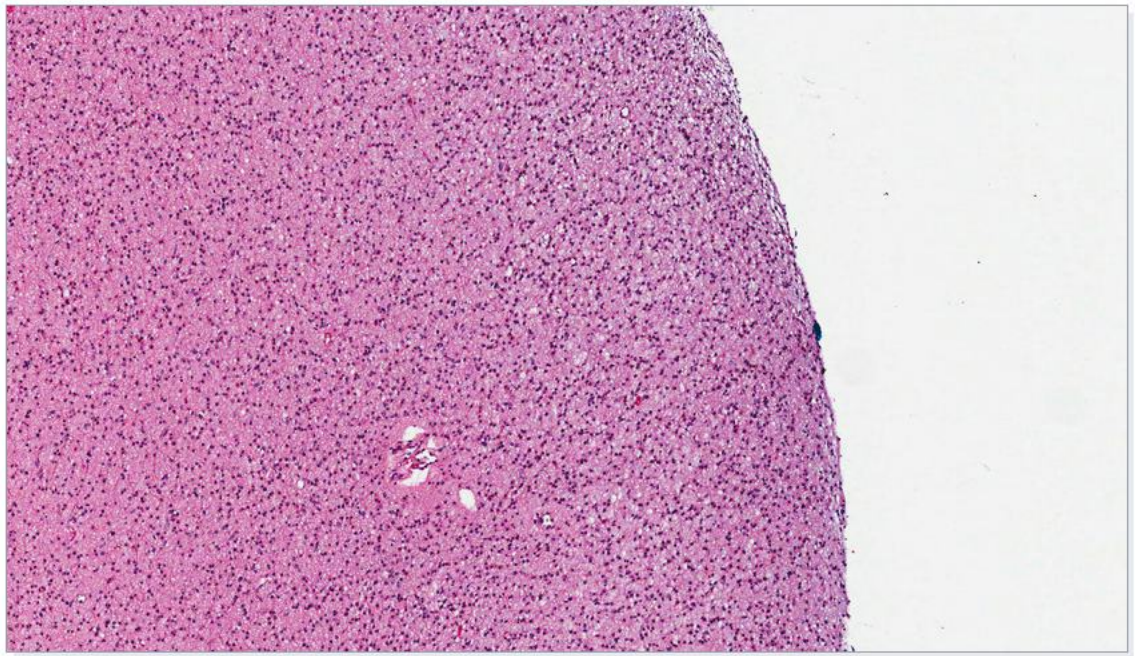
1. Shown for studies with H&E slide images.
2. 1,234 slides from the 395 selected patients. Filters apply.
3. One row per slide, with pathology annotations: tumor cells, tumor nuclei, necrosis (%).
4. Open a slide in the image viewer (next slide).

Link to this page

Slide Microscopy: viewing an H&E slide



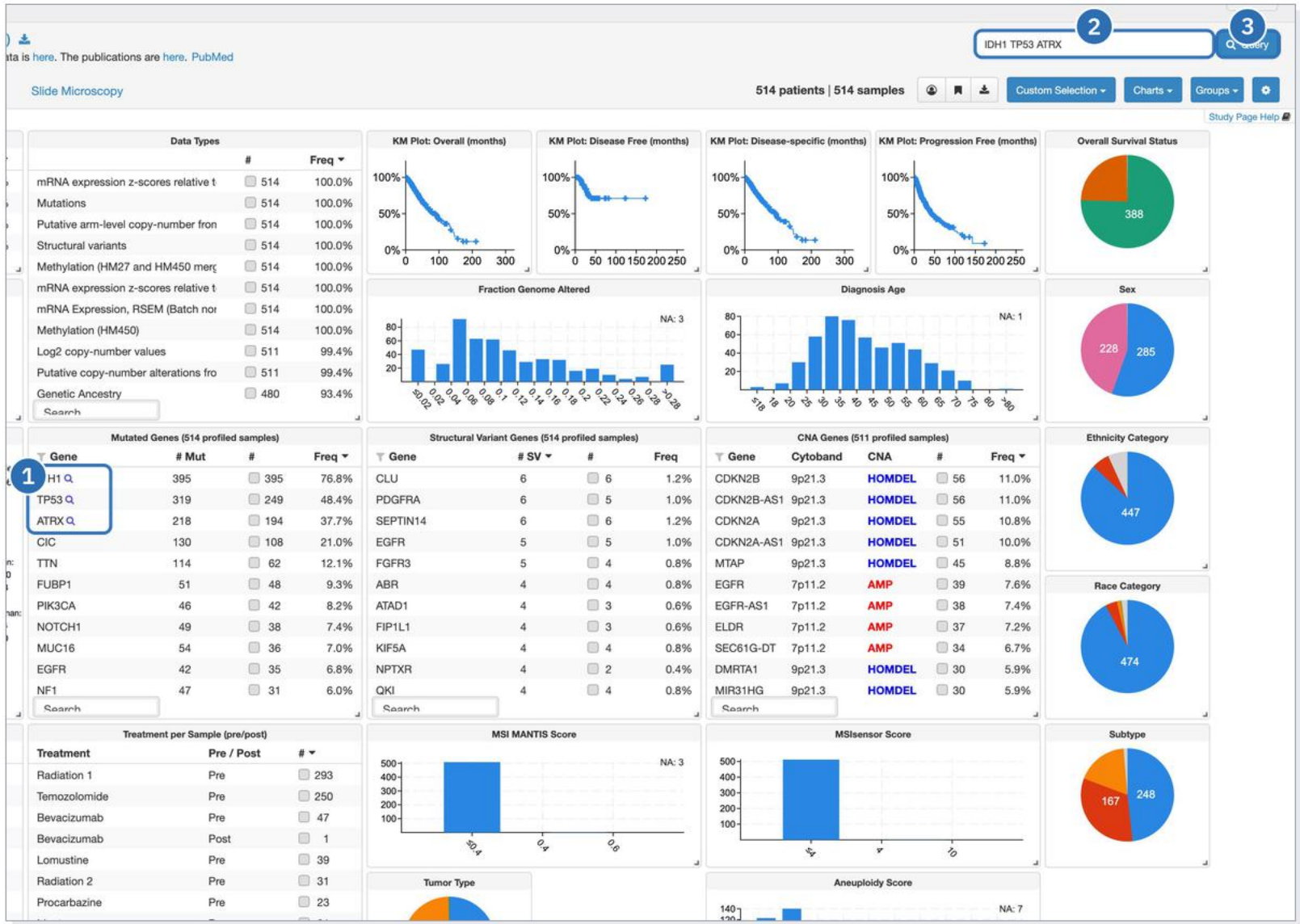
Open in new window shows the slide in the NCI Imaging Data Commons viewer. Here: TCGA-CS-4938, FFPE H&E; the patient's other slides are listed on the left.



Scroll to zoom in down to the cells. The overview map (bottom left of the viewer) shows where you are on the slide.

Link to this page

Study view: run a query



1. Click a gene in any genomic table to add it to the query. Apply filters first to query only the selected samples.
2. Or type gene symbols here.
3. Then click Query. See Tutorial 2: Single Study Query.

Link to this page



Questions?

Next: Tutorial 2, Single Study Query

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